BrickPlan 2.0: Certified Resource and Inspection Planning with Visual Assembly Observations

Benchmark research release

October 8, 2026

Abstract

Long assembly sequences need not require substantial planning. An exhaustive audit of our preceding benchmark finds that a static blocker-closure formula exactly recovers all 1,802,728 stored state–goal costs. We redesign the task around a finite external rack and persistent off-assembly inspection obligations, while retaining original source poses and qualified insertion paths. The resulting bank contains 69 interactive episodes, 196 diagnostic responses and 59 matched pairs across 6 profiles from 5 source kits. Directed exact oracles certify every retained product state and transition. The static closure bound now underestimates completion cost by up to 18 operations; identical assemblies can require disjoint optimal next actions when their visible inspection ledgers differ. Across 207 repeated task-runs, closure greedy completes 189 but is optimal in only 91; capacity-aware dispatch completes 170, and closure A* with 100 expansions completes 162. Full closure A* solves every task optimally with at most 2,555 expansions. This is evidence of nontrivial resource scheduling under a declared abstraction, not general computational hardness. We provide matched image and structured interfaces, conditional-cost and recovery metrics, authoritative transcript replay, and budget-accounted external-policy collection. VLM and human experiments remain unmeasured.

1 Introduction

A benchmark can contain visually complex objects, long solutions and many temporary removals while still admitting a short analytic rule. Such a benchmark may test relation extraction, but its solution length alone does not establish a planning challenge. BrickPlan 2.0 begins with an explicit falsification of the earlier difficulty claim and a mechanism-level repair.

Consider a technician who must restore an assembly and inspect selected parts outside it. A removed part occupies a rack holder. If the rack is full, another piece may have to be installed temporarily before a necessary removal is possible. Inspection persists after reattachment: two visually identical assemblies can have different unfinished obligations. The final shape, current occupancy and static access relations no longer determine the complete mission state.

Table 1: Planned empirical slots. No row has measured results. Exact model versions, providers and participant cohorts must be registered before collection.

Condition	Measurement status
VLM, image+ID, stepwise actions	Not evaluated
Same VLM, image+ID, explicit plan	Not evaluated
Same VLM, structured input	Not evaluated
Human, image+ID	Not evaluated
Human, structured input	Not evaluated

Our contribution is an executable diagnostic contract: a new resource and inspection state space with exact labels, paired tests that isolate intermediate order and mission history, and metrics separating immediate illegality from added future cost and delayed recovery. We make the selection process and simple-control performance explicit. Neither large scene size nor a failed weak baseline substitutes for this evidence.

Relation to prior assembly work. Break and Make introduces interactive disassembly and reconstruction in LTRON [1]. InstructioNet builds an explicit visual instruction memory to support reconstruction [2]. These works already establish interactive structural understanding and long assembly sequences. BrickPlan uses a narrower known-site action space to obtain exact counterfactual costs and event-level error labels. Our visible inspection ledger is a declared task variable, not a claim of latent episodic memory. Resource limits, shortest-path search and admissible heuristics are established planning tools [3]; our contribution is their audited integration with source-qualified visual assembly diagnostics.

2 Experimental design and reporting contract

All conditions share task assignments, identity seeds, operation limits and camera options. Each manifest records participant/model identity, condition, version, prompt

and configuration hashes, provider, temperature, reasoning setting, input/output limits, request cap and accounting mode. We recommend an initial format-and-cost pilot followed by preregistered collection; the release does not report simulated human or VLM scores.

Primary outcomes are completion over all assigned episodes, optimal completion, successful regret with coverage, and conditional cost added over all observed decisions. Diagnostic outcomes cover next-action choice, continuation failure localization, paired consistency, diagnosis and recovery. Software search sees the private relation graph and is reported separately from both public tracks.

Roadster and kart share kit 31027. Development groups are 31027 and 31028; 10156, 31009 and 5867 carry held-out-source labels. The generator has used all five groups' geometry and solver results to select challenge instances. This is therefore a public challenge partition, not a blind held-out test. Seeds permute identities and tie orders; they do not create independent structures. Results require source-group summaries and explicit missing assignments. Five groups do not support strong population-level claims.

3 Why the preceding bank was simple

Let s and g be sets of installed registered sites. Initialize a removal set with occupied unwanted sites and occupied blockers of missing target sites. Repeatedly add occupied blockers of every site already in that set, obtaining $R(s, g)$. The static quantity

$$h_{\text{old}}(s, g) = |s \triangle g| + 2|R(s, g) \cap g| \tag{1}$$

counts required occupancy changes and detach–reattach pairs for occupied target pieces that must be removed.

An exhaustive audit compares this formula with all reachable labels in every stored goal table of version 1.2: 70,060 roadster labels, 312,228 kart labels and 1,420,440 airframe labels. Every comparison is equal. Computing Eq. 1 needs static relations but no state-space search or cost table. This is a finite-bank equality result, not a theorem for all assembly geometries. It is stronger than observing that a rule solves the 15 selected episodes.

Unlimited temporary storage permitted an all-removals-then-attachments strategy. Occupancy encoded the entire state; no completed work survived outside that encoding. Supplied-plan failure localization could measure execution checking without measuring action selection. Hamming-only lookahead controls substantially understated what static relations could already provide. These are task-design issues rather than an insufficient number of distractors.

4 Certified workcell and mission state

4.1 Source-qualified physical transitions

Each registered physical piece P_i fits one original source site S_i; even identical part types are not interchangeable in this protocol. Unregistered fixed parts form one stud-connected clamped subassembly. Every installed piece must remain stud-connected to it. Attach/detach operations follow a qualified straight translation at the original source orientation, with current blockers and endpoint connectivity determining legality. No goal or future-cost table participates in execution.

The geometry qualification uses closed collision proxies, a 0.25 LDU surface inset and a 0.002 LDU guard inherited from the predecessor. Endpoint tests and continuous straight-path tests reject intersections or unresolved geometry. Interval clearance uses a translation-distance bound; the source display meshes remain original LDraw triangles. The qualification is numerical evidence within this proxy/path model, not measured contact mechanics.

We retain the three earlier profiles and qualify truck, cottage roof and sports-car profiles without changing source transforms. A pizza-shop candidate fails endpoint qualification and is excluded. Larger scenes do not automatically supply more movable sites: the cottage profile has 18 registered sites but only 17 move in its full-assembly component; the sports-car profile has 22 registered sites but only 10 move. The latter remains an easy control.

4.2 Finite rack and persistent work orders

Let the full mission state be $x = (s, z)$, where z is the set of completed inspection jobs. The mission specifies target g, rack capacity K, and declared job set J. Every detached registered piece occupies one universal oversized holder:

$$q(s) = n - |s| \le K.$$

A legal detach must preserve this inequality. An `audit` of i is legal exactly when $i \in J \setminus z$ and $i \notin s$; it changes z to $z \cup \{i\}$. Completion requires $(s, z) = (g, J)$ and an explicit `finish`. The ledger is visible and persists after reattachment. An eligible audit never competes for a hidden resource or outcome.

Each structural attempt or audit attempt costs one operation, including rejections. Valid camera acquisitions, diagnoses and finish cost zero. Budgets depend only on visible size: $6n + 2|J|$ operation attempts, 12 additional views and 12 diagnoses. Spending the final operation without finishing terminates by budget. Detached-piece transfers to rack holders are abstracted; we do not certify a physical rack layout or hand trajectory. Executable installed states retain the source poses.

Table 2: Source census. "Parts" counts the retained source subassembly, not necessarily the complete source kit. "Move" counts registered sites that can move in the full-assembly component with unlimited rack capacity. "Rack" is the finite challenge capacity; D1 controls may use a larger rack.

Profile	Kit	Parts	Reg.	Move	Rack	Episodes	Max gap
Roadster body	31027	37	17	17	9	14	18
Kart deck	31027	25	18	18	5	14	10
Airframe	31028	24	21	21	8	14	4
Truck cab and roof	10156	76	18	18	12	10	6
House roof assembly	31009	221	18	17	5	14	4
Sports car upper body	5867	149	22	10	8	3	0

Table 3: Operational tiers. A* counts use the closure heuristic and seed 0. Tiers encode mechanisms, not a universal monotone hardness ranking.

Tier	Episodes	C^* range	Gap range	A* max
D1	13	10–33	0–0	33
D2	16	16–42	2–18	2555
D3	34	9–40	0–10	2185
D4	6	33–42	6–10	2131

4.3 Exact costs and a strong static bound

We compress the reachable physical component to an indexed graph and take its product with the $2^{|J|}$ inspection flags. Goal-rooted reverse BFS handles physical edges in both directions; a reverse audit clears a flag only at a state where its piece is detached. Let $C^*(x)$ be exact remaining operations. Every table is exhaustively checked for terminal labels, Bellman equality, runtime-edge agreement, physical inverses and the lower bound below. The 35 retained tables contain 1,420,363 state entries and 11,230,303 directed transition entries. These totals include repeated physical states under different missions; they are not independent examples.

Extend R to include all occupied unfinished inspection pieces before taking occupied-blocker closure. A resource-relaxed bound is

$$h(x) = |s \triangle g| + 2|R(x) \cap g| + |J \setminus z|. \tag{2}$$

It counts unavoidable operations while dropping rack and support scheduling constraints. Every retained label satisfies $h(x) \le C^*(x)$. The gap $C^*(x) - h(x)$ reaches 18. For the roadster task in Fig. 1, the start has nine rack occupants, the bound is 24 and the optimum is 42; its unique optimal first operation is an attachment. This supplies a direct counterexample to reusing the old exact-cost formula.

5 Challenge construction and matched diagnostics

D1 retains unlimited-rack and measured easy inspection controls. D2 selects finite-rack reconfiguration with gap at least two. D3 includes inspection/resource missions and certified history or changed-work contrasts; paired members need not individually have a positive gap. D4 requires at least two jobs, gap at least four and more than 100 closure-A* expansions under seed 0. Instances are selected from exact rankings with start-state diversity constraints. These labels are explicitly conditioned on the chosen mechanism and algorithm. Some D2 instances require more expansions than D4 instances.

The bank has 98 next-action and 98 continuation responses. Next-action checkpoints include legal actions that differ in exact conditional cost. Every cost-minimizing offered action receives credit, including ties. Forty-nine continuation pairs preserve the initial state, first and last operations, length and action multiset. Swapping interior operations creates the contrast between a legal plan and its first execution conflict. A classifier restricted to the invariant features cannot correctly label both members: its response accuracy is at most 50%, and its joint pair accuracy is zero. The release tests this particular shortcut, not all possible shortcuts or balanced failure positions.

Five history pairs share assembly, physical target and work-order set but differ in completed flags. Five future-work pairs share assembly and physical target but change the requested inspection piece. Each pair has disjoint optimal initial action sets, and neither set contains an audit. This avoids reducing the contrast to "audit the only available job." The changed information is public; hidden memory is not required. These ten pairs are episode pairs; the other 49 are diagnostic pairs.

6 Visual observations and collection

The image+ID track supplies five PNG panels: current assembly, target, all-site reference, registered-piece catalog and occupied rack. Public text includes registered P/S mapping, rack counts, work orders and budgets. It excludes source matrices, installed-site flags, blocker/support graphs, legal-action lists, depth buffers, costs, tiers and pair identities. The catalog and fixed registration deliberately simplify correspondence. Rack pixels also expose which physical pieces are outside the assembly.

This track still requires interpreting shape and access, but is not a test of wholly implicit object identity.

The structured track adds matrices, types, colors, paths and occupancy. It also includes the same rasters, without the private relation graph. Humans and agents use the same seven fixed cameras and exploded reference option. Explosion changes only the reference diagram, never executable current or target states. Each acquired view is saved at its revision; revisiting stored pixels is free and does not change executable state. Free 3D orbit, zoom and explosion are available only after a practice episode ends.

The local server fixes condition, seed, assignments and run identity. Study starts are single-use and probe answers freeze on first acceptance. Revisioned requests are idempotent across restart; persistence is atomic with in-memory rollback on failure. Administrative collection replays every episode and probe request and rejects mismatched identities, answers and view histories. Missing assignments remain in the census. Static routes form an allowlist; private bank, geometry and cost files are not served in study mode.

The external adapter consumes public JSON and must pass every raster to the provider as an image, including retained views. Before invocation, the runner reserves the maximum declared request tokens and cost. Returned usage must fit the reservation; missing or uncertain usage stops collection. These limits depend on a trusted adapter/provider report, not independent billing enforcement. A separate "unavailable" mode enforces only request counts and process timeouts. An external working directory alone is not a sandbox; defensible studies place the policy on a host without evaluator files.

7 Seven measurement dimensions

1. Avoidable-cost and budget-loss choices. For an operation of cost c_t with successor x_{t+1}, define

$$\epsilon_t = c_t + C^*(x_{t+1}) - C^*(x_t) \geq 0.$$

A legal decision enters the avoidable-choice denominator only when both zero-excess and positive-excess legal choices exist. Its numerator counts chosen positive excess. This detects errors even when rework was already mandatory. Budget loss instead asks whether a previously budget-feasible mission becomes infeasible after expenditure, using its own mixed-choice denominator. Invalid attempts are separate. All retained legal states remain solvable. Consequently, the genuine irreversible-dead-end denominator is zero and its rate is undefined, not a measured zero.

2. Lookahead diagnostics. First-conflict localization scores the supplied plan; it does not measure internal search depth. For action selection, independently compute $H_0 = h$ and $H_d(x) = \min_a[1 + H_{d-1}(T(x, a))]$, with the terminal absorbing. At depths 1, 2, 4 and 8, report whether any or all minimizing actions are exactly optimal, plus uniform-tie correctness. These are algorithmic choice bounds. Correlated checkpoints and challenge-conditioned selection preclude interpreting them as a population distribution of model reasoning depth. Depth-one minima already contain an optimal action at every selected checkpoint; the remaining uncertainty is which tied action to select.

3. Plan survival. The first positive ϵ_t is a permanent historical loss even after recovery. At attempt t, the assigned set contains episodes with declared budget at least t. Successful completions are absorbing survivors; unfinished short traces and missing episodes are censored. Report known-survivor and possible-survivor fractions as bounds, together with the changing risk set. This is a descriptive assignment-level curve.

4. Regret. Successful regret is $R = T - C^*(x_0)$, where T counts attempted operations. Failures have null successful regret, explicit cost spent and remaining cost. The telescoping identity $\sum_t \epsilon_t = T + C^*(x_T) - C^*(x_0)$ holds on all observed traces. Writing $u = |J \setminus z_0|$ and $\Delta = |g| - |s_0|$ gives

$$D^* = \frac{C^*(x_0) - u - \Delta}{2}, \qquad R = 2(D - D^*) + N_{\text{invalid}}.$$

The inspection term is necessary: every unfinished job must be audited once, while duplicate audits are rejected.

5. Recovery. Every legal positive-excess action starts an event, including overlapping mistakes. For its pre-error checkpoint b, define the optimal corridor

$$\mathcal{C}_b = \{x : d(b, x) + C^*(x) = C^*(b)\}.$$

Recovery is the first return to this set. A physical inverse provides a one-operation restoration path; an eligible audit cannot add excess under this contract. We retain diagnosis sign/lag, restoration attempts, terminated-unrestored events and censored events. The goal belongs to every corridor, so eventual completion counts as restoration; lag and efficiency distinguish prompt correction from this late outcome.

6. Replanning. Restored-event efficiency is minimum recovery operations divided by actual operations to restoration, here $1/k$. A true positive diagnosis also opens a suffix record: remaining optimum, operations spent afterwards, completion and successful residual regret. Diagnoses ask about new legal cost additions since the last diagnosis. Report all events and coverage; diagnosis accuracy alone is susceptible to always-negative answers.

7. Counterfactual consistency. History and changed-work pairs require an optimal first attempted physical operation in both members, with immutable randomized registration. Continuation pairs require correct answers in both members. A changed answer alone is insufficient; missing members receive no joint credit. These are pair-specific joint scores, not independent task successes.

8 Measured software evidence

Closure greedy chooses minimum h successors, breaking ties with visited states and randomized site rank. Capacity dispatch prioritizes available audits, mandatory removals, safe target placements and temporary staging, with a visited-state escape rule. Both are resource-aware no-search controls, stronger than blindly applying the old all-removals policy. Their 91 optimal completions each out of 207 demonstrate that cost-sensitive evaluation exposes substantial errors even when greedy completion is high.

A* uses graph reopening, a $(g + h, h)$ priority and declared expansion caps; it never reads an oracle table. Increasing the cap from 100 to 1,000 raises completion from 162 to 183; full closure A* achieves 207 optimal completions with median 30 and maximum 2,555 expansions. The same search with the weaker occupancy-plus-pending-audit heuristic also solves all tasks, but has median 3,706 and maximum 75,454 expansions. This difference demonstrates why a weak heuristic can inflate an apparent planning challenge. Full closure search is computationally modest.

Dropping inspections completes only 84 missions despite reaching its relaxed physical target in every run. Planning with unlimited rack completes 62; 145 runs encounter executor rejection. These are deliberate model-mismatch ablations, not evidence that the unmodified policies are competent resource planners. No policies are forced to select known traps. Timings are archived local implementation measurements with shared caches, not calibrated cross-platform performance estimates.

9 Validation, limitations and use

Exhaustive oracle certificates are supplemented by independent runtime edge checks, replay of all 69 episode witnesses and 196 probe transcripts, regret identities, overlapping recovery events, missing-assignment censoring and tamper rejection. Ten protocol tests cover restart, idempotence, rollback, usage accounting, study isolation and a real offline adapter-to-HTTP-to-collection round trip. Ten Firefox checks exercise rasters, saved views, action guards, replay, 3D review, frozen probes, persistence and mobile layout. These checks validate software behavior; they are not human usability or hardware construction experiments.

The upgraded bank defeats a demonstrated exact static-cost shortcut and introduces measurable mission-state dependence, but it remains small, public and deterministic. All states are finite and all retained legal actions preserve solvability. Inspection is deterministic and public; holder capacity abstracts workspace resources. We do not model free poses, rotation, simultaneous subassembly motion, gravity, clutch forces, hand access or rack transfer geometry. A source drawing or a collision proxy does not validate these omitted phenomena.

The current evidence supports a reproducible research instrument and algorithm-conditioned challenge bank. It does not establish visual-model shortsightedness, a human–model gap, robustness to all shortcuts or publication readiness. The next empirical requirement is a preregistered image-versus-structured model/human study, followed by independently acquired source families and additional mechanisms. The release preserves earlier versions and provides its exact scope, negative findings and reproducible artifacts so those claims can be tested.

References

[1] Aaron Walsman, Muru Zhang, Klemen Kotar, Karthik Desingh, Ali Farhadi, and Dieter Fox. Break and Make: Interactive Structural Understanding Using LEGO Bricks. ECCV, 2022. https://arxiv.org/abs/2207.13738.

[2] Aaron Walsman, Muru Zhang, Adam Fishman, Ali Farhadi, and Dieter Fox. Learning to Build by Building Your Own Instructions. 2024. https://arxiv.org/abs/2410.01111.

[3] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A Formal Basis for the Heuristic Determination of Minimum Cost Paths. IEEE Transactions on Systems Science and Cybernetics, 4(2):100–107, 1968.

Table 4: Executed software controls. Non-oracle controls run three tie-order seeds over the same 69 episodes; the oracle runs once. "Optimal" counts successful zero-regret runs, "Regret" is the mean over successes, and "Max exp." counts expanded A* states. All controls use privileged structural information; these are not public-track model results.

Control	Runs	Success	Optimal	Regret	Max exp.
Exact oracle	69	69	69	0.00	—
Closure greedy	207	189	91	12.39	—
Capacity dispatch	207	170	91	12.14	—
Closure A* (100)	207	162	162	0.00	100
Closure A* (1,000)	207	183	183	0.00	1,000
Closure A* (full)	207	207	207	0.00	2,555
Hamming A* (full)	207	207	207	0.00	75,454
Legal random	207	1	0	90.00	—
Drop inspections	207	84	84	0.00	—
Ignore rack limit	207	62	62	0.00	—

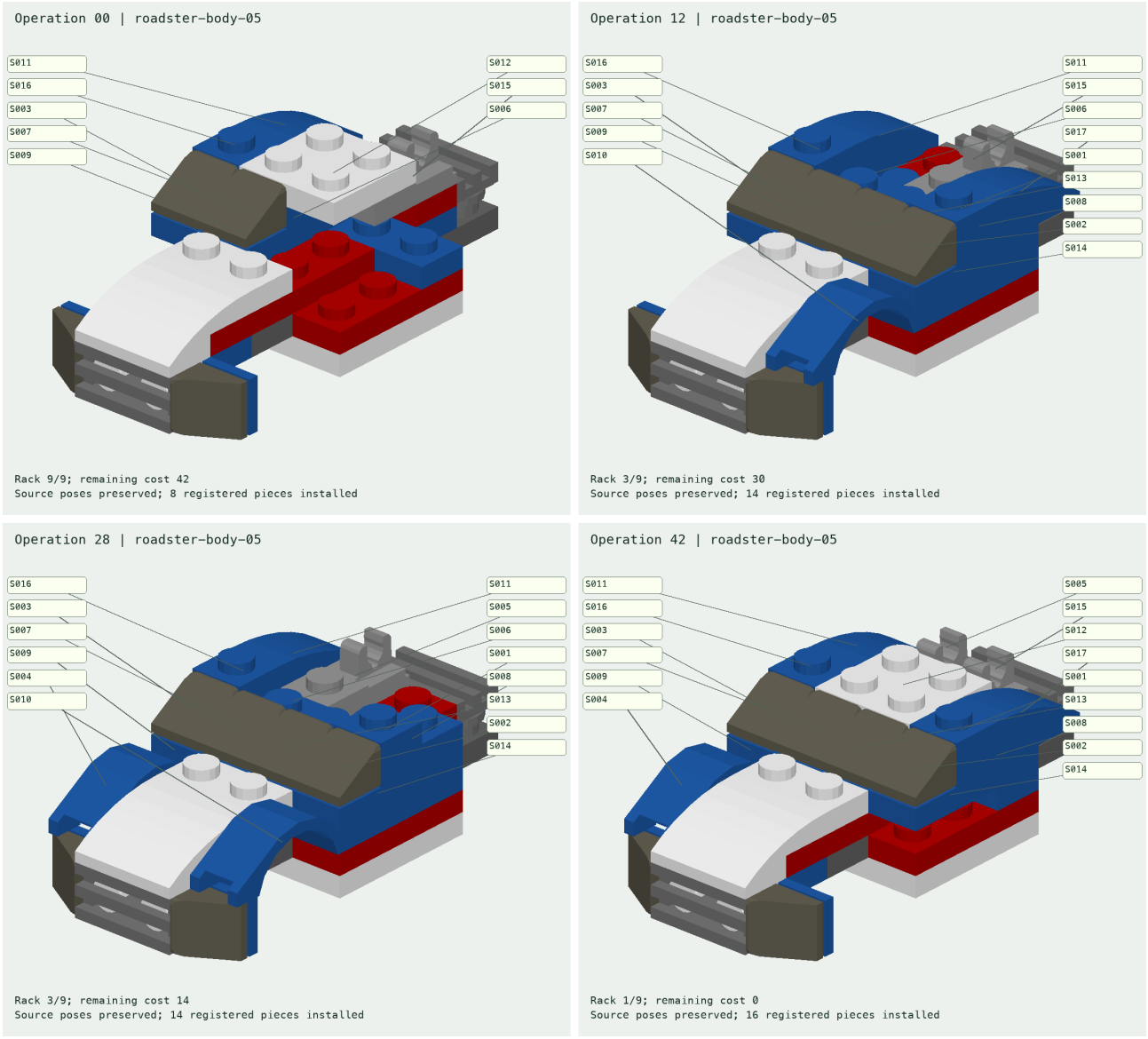


Figure 1: Authoritative replay of roadster-body-05, identity seed 31. All four assemblies retain source poses and stud connectivity. The finite rack requires temporary installation and repeated access restoration: $h = 24$ versus $C^* = 42$. Labels identify source sites. Detached pieces occupy the external rack abstraction and are not depicted as suspended installed pieces. Frame states and image hashes are supplied.

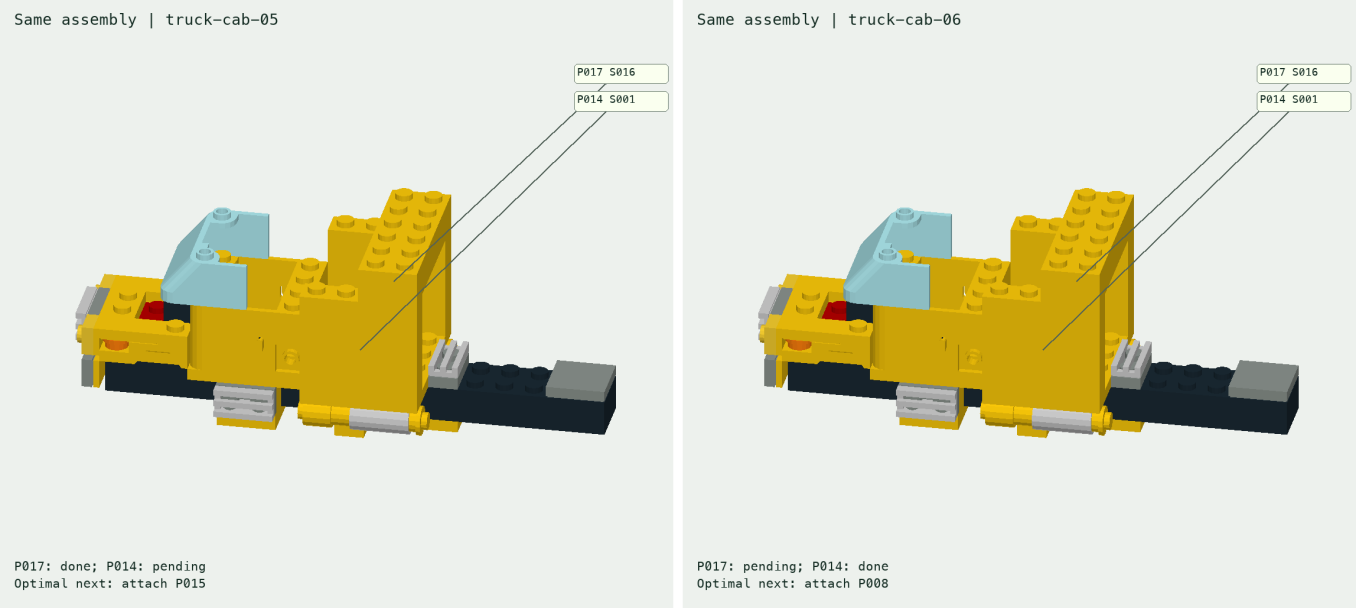


Figure 2: Truck history pair with identical assembly occupancy and different visible completion ledgers. The certified optimal next-action sets are disjoint. Figure labels expose evaluator answers for explanation; study observations omit those answers. All P/S identifiers follow seed 31.

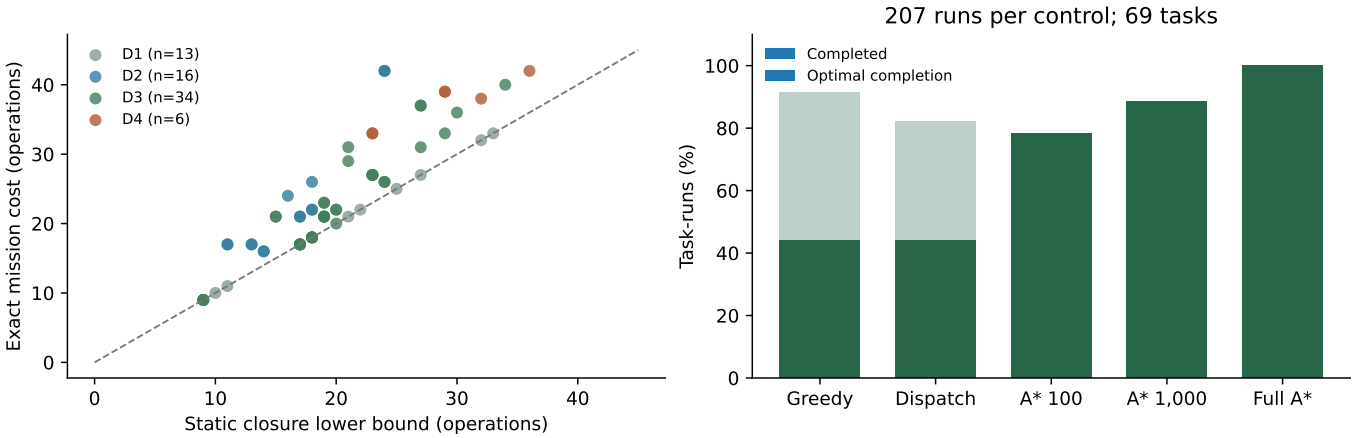


Figure 3: Left: exact cost versus the strong static bound for all 69 episodes; coincident points may overlap. Right: actual completion and optimal completion for structural controls. The tiers and tasks were selected using mechanism and solver evidence, so these are descriptive challenge-bank measurements.

A Exact oracle and audit details

Static audit reproducibility. The archived `data/predecessor-static-audit.json` contains one row per preceding profile. The original research audit enumerates every nonnegative label in every stored goal table, computes occupied-blocker closure and compares Eq. 1 with the label. This establishes equality only over that finite census. It neither proves the formula exact on arbitrary goals nor reconstructs a plan from visual observations.

Admissibility of the new bound. An initially unwanted occupied piece must be removed. A missing target piece must be inserted, requiring removal of every occupied blocker along its fixed insertion path. Every occupied unfinished inspection piece must be removed to a rack holder. Each required removal recursively requires removal of its occupied path blockers. Therefore every occupied target piece in the resulting closure must be detached and reattached at least once. Initially mismatched sites need at least one operation each, and each unfinished audit costs one. These disjoint operation counts give Eq. 2. Connectivity and rack scheduling can add operations but cannot invalidate the lower bound.

Bellman certificates. For a nonterminal product node x, the verifier checks $C^*(x) = \min_{a \in A(x)}(1 + C^*(T(x, a)))$ and $C^*(g, J) = 0$. Each physical graph edge is compared with the runtime's independent transition enumeration, and its inverse is executed. Audit edges are also replayed through the mission runtime. Every state in every retained table reaches the terminal. Counts across tables repeat physical states under different goals/job sets and must not be presented as the number of independent tasks.

Why no genuine irreversible dead ends exist. The physical graph is reversible within its capacity-constrained component. All admitted job pieces can be exposed in that component. Completing a job adds a persistent bit and never removes an available physical action. Thus any retained legal successor can still reach the goal and finish remaining audits. A legal choice can add cost, consume scarce budget or delay inspection, but it cannot cause permanent geometric unreachability. This is why the unreachable-choice metric is undefined in this release.

B Heuristic and control diagnostics

Depth	Choices	All optimal	Any optimal	Uniform tie
1	98	49	98	74.7%
2	98	61	98	78.9%
4	98	49	98	70.8%
8	98	66	98	81.8%

The table evaluates the 98 selected next-action probes. "All optimal" is a pessimistic tie bound; "Any optimal" is the optimistic bound. The depth-one minimum already includes an optimal action at every selected checkpoint. Consequently, these probes do not show that shallow reasoning must exclude the correct action. They expose ambiguity among tied choices. Uniform tie correctness is a theoretical average over the minimizing set, not a sampled policy run. Deeper search can change ties nonmonotonically; the measured depth-four result is worse than depth two and is retained without relabeling. No "required internal lookahead depth" is inferred.

Control	D1	D2	D3	D4
Closure greedy	39/39	42/48	96/102	12/18
Capacity dispatch	39/39	42/48	83/102	6/18
Closure A* (100)	39/39	39/48	84/102	0/18
Closure A* (1,000)	39/39	42/48	93/102	9/18
Closure A* (full)	39/39	48/48	102/102	18/18
Hamming A* (full)	39/39	48/48	102/102	18/18

Counts are task-runs, with three repeated tie-order seeds. D4 is explicitly selected using the 100-expansion closure-A* criterion, so its zero completion at that cap is partly a selection property. D2 can exceed D4 in search effort. Some source groups remain easy: geometry size, tier and solution length should not be pooled into a single unqualified hardness claim.

All raw plans, final states, node expansions, elapsed times, rejection codes and per-run regrets are archived in `data/software-control-records.json.gz`. Aggregates are in `data/software-controls.json`; `controls.py` recomputes them. The 500,000 expansion cap is an upper resource limit, not the number of expansions consumed.

Oracle table loading time is not comparable to online A* time. Local timings share caches and lack statistical calibration.

C Metric denominators and event semantics

Output	Denominator / eligibility	Missing and termination treatment
Completion	All assigned episodes	Unstarted and unfinished episodes are not successful
Avoidable-cost choice	Legal decisions with both zero- and positive-excess alternatives	Invalid attempts excluded; their cost remains in total excess
Budget-loss choice	Legal decisions with viable start budget and both viable/nonviable alternatives	Other budget losses remain in decision records
True unreachability	No eligible choices in retained components	Undefined, not zero evidence
Successful mean regret	Successfully finished episodes	Null for failures; always accompany with success coverage
Recovery bounds	Every legal positive-excess event	Lower: restored/all; upper adds only censored events
Diagnosis accuracy	Accepted diagnosis requests	Unsampled intervals are not inferred correct
Pair consistency	Fully assigned pairs	Missing member prevents joint credit
Probe correctness	All assigned probes of that kind	Missing or invalid answer gets no credit

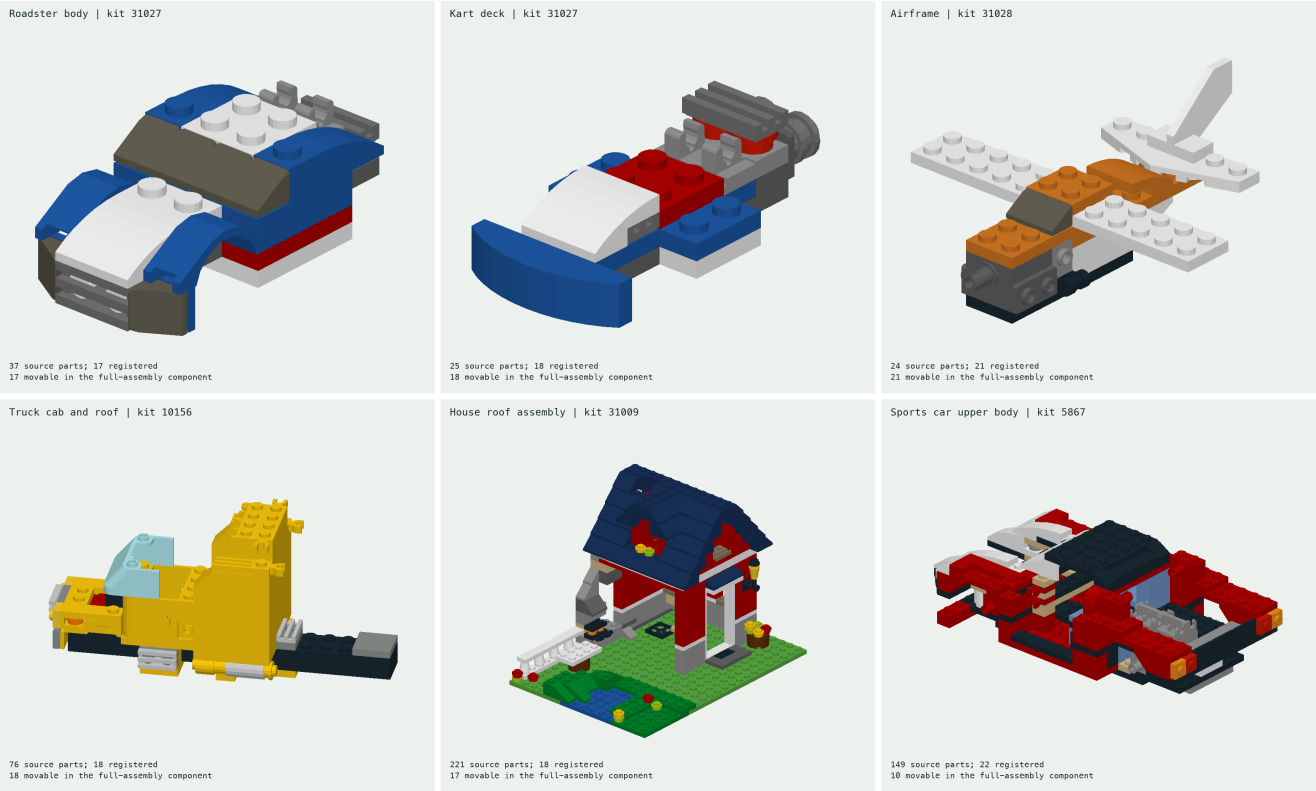
Recovery corridor computation. Starting from checkpoint b, traverse only edges satisfying $C^*(T(x,a)) = C^*(x) - 1$. The reached states are exactly $\mathcal{C}_b$: every such traversal is an optimal prefix, and every state on an optimal completion has one. Corridor membership can therefore be checked without exploring irrelevant detours. Each new positive-excess action starts a separate event. Later physical steps update every active event, so a second error cannot overwrite the first. Rejected operations consume time but cannot change corridor membership. Terminal-unrestored events are known failures to restore; unfinished events are censored.

Inspection and regret conservation. Successful attaches and detaches satisfy $A - D = \Delta$. Every successful trace contains exactly u accepted audits; duplicate audits are invalid and do not change the ledger. Hence $T = A + D + u + N_{\text{invalid}} = 2D + \Delta + u + N_{\text{invalid}}$. An optimal trace contains no rejection, giving $C^*(x_0) = 2D^* + \Delta + u$ and the regret identity in the main text. Failed finish attempts and rejected camera/diagnosis requests are charged when the runtime classifies them as invalid operation attempts. Network-level malformed envelopes and stale revisions are rejected before mutation and do not become action events.

Diagnosis and suffix reports. A diagnosis refers to legal avoidable-cost events since the previous diagnosis or start, rather than whether the current assembly merely needs rework. The first later diagnosis is attached to each previously undiagnosed event, with its sign and attempt lag. Only true positive diagnoses open replanning suffix records. The report retains all raw events so event coverage, false alarms and repeated reports can be inspected separately. An always-negative or selectively queried strategy can make aggregate diagnosis accuracy misleading; it must be read with error-event detection.

Survival interpretation. The default curve is pooled over assigned task budgets. It is a descriptive record with explicit denominators at each operation index, not a Kaplan–Meier population estimate. Missing and short unfinished traces contribute unknown outcomes; restoration never erases the first cost loss. Compare conditions only on matched assignments and report the task/source composition. An experiment requiring source-stratified curves should register and report those strata before pooling.

D Sources, geometry and visual provenance



All six full source subassemblies are shown without exploded transforms. The fixed part count can dominate the visual scene and does not increase the number of decision variables. Roadster and kart are grouped together by source kit. The cottage and sports car retain mechanically locked registered sites; their active size is reported explicitly.

The flattened display atlas contains 94 distinct part meshes from 336 locked source files. `data/render-provenance.json` records every source hash and the mesh-atlas digest. The generated figures use the same triangle raster as the observation interface, at publication resolution. `figures/provenance.json` binds task states, seed, mesh digest and figure hashes. The plot uses measured task/control data. No generated illustrative substitute meshes or model-output images are used.

Source models are attributed to Merlijn Wissink (31027 and 31028), Robert Paciorek (10156), Stefan Frenz (31009), and Takeshi Takahashi (5867), with the exact OMR URLs and licenses in `ASSET-LICENSES.json`. Part-level authors and notices are preserved there separately. The source models use CC BY 2.0; part notices identify their applicable CC BY licenses. Original LDraw triangles, source matrices and registered IDs remain traceable. Geometry proxies are used only for qualification, not as substituted display shapes.

E Protocol, trust boundaries and acceptance evidence

The full HTTP and external-adapter schemas are specified in `PROTOCOL.md`. Study collections bind run, participant, condition, seed, assignments and bank hash. An evaluator replays all mutation requests, recomputes outcomes, verifies view histories and normalizes permuted candidate indices before scoring. A request ID reused with changed content or stale revision is rejected. Identical retries return their original acknowledgment without repeating operations.

Study mode excludes score/replay/free-3D endpoints and private static assets. Practice exposes those only for ended episodes. Neither local files nor export digests form a cryptographic guarantee against an administrator editing both data and code. Records must remain on the evaluator host. The loopback server is a single-participant development service; remote deployments require per-participant isolation and an authenticated gateway.

The external process adapter has no built-in provider call or paid model. Reported-usage mode reserves the declared maximum per-call input/output tokens and currency before invocation. The adapter must account for all images and context, enforce limits and attest that it did so. Incomplete usage, timeout or malformed output leaves a durable failed or unknown call and stops automatic continuation. Unknown provider usage cannot be silently cleared by

restarting. Unavailable-accounting mode reports tokens and currency as unavailable, not as zero. Every call has full input/output, identity and configuration hashes, timestamps/latency, usage, request IDs and error records.

Acceptance layer	Evidence
Exact domain/oracle	35 exhaustive tables; runtime edges, Bellman equalities, lower bounds and physical inverses
Semantic evaluation	69 episode replays; 196 probe transcripts; cost/recovery/censoring identities and tamper rejection
HTTP / adapter	10 tests covering restart, idempotence, view limits, rollback, private-route isolation, usage budgets and offline end-to-end collection
Browser	10 Firefox checks; source pixels, saved views, action guards, 3D orbit/explosion, frozen probes, reload and mobile layout
Manuscript and release	Generated table/data consistency, figure provenance, predecessor hash audit and release manifest

Reference-action software acceptance is not participant performance. Browser screenshots and test outputs are archived under `tem/brickplan-v2/validation/`. The current manuscript uses a generic two-column research layout; conference-template formatting, full contemporary related-work coverage and empirical model/human results remain submission work.

F Complete episode census

"Jobs" is the total declared inspection count, "Done" is its initially completed subset, and A* expansions use seed 0 and the closure heuristic. Internal IDs below are evaluator documentation; participant catalogs use opaque identifiers. Exact start/goal masks, ordered jobs, optimal actions, witnesses and oracle paths are in the private task bank.

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
roadster-body-01	D1	17	0	0	17	17	17
roadster-body-02	D1	17	0	0	32	32	32
roadster-body-03	D2	9	0	0	27	23	33
roadster-body-04	D2	9	0	0	27	23	38
roadster-body-05	D2	9	0	0	42	24	2555
roadster-body-06	D2	9	0	0	42	24	2555
roadster-body-07	D3	9	2	0	37	27	2185
roadster-body-08	D3	9	2	0	37	27	2139
roadster-body-09	D3	9	2	1	20	20	21
roadster-body-10	D3	9	2	1	18	18	18
roadster-body-11	D3	9	1	0	18	18	18
roadster-body-12	D3	9	1	0	18	18	18
roadster-body-13	D4	9	4	0	39	29	2131
roadster-body-14	D4	9	4	0	39	29	2129
kart-deck-01	D1	18	0	0	18	18	18
kart-deck-02	D1	18	0	0	25	25	25
kart-deck-03	D2	5	0	0	17	11	32
kart-deck-04	D2	5	0	0	17	11	32
kart-deck-05	D2	5	0	0	26	18	119
kart-deck-06	D2	5	0	0	24	16	90
kart-deck-07	D3	5	2	0	31	21	121
kart-deck-08	D3	5	2	0	29	21	218
kart-deck-09	D3	5	2	1	21	15	68
kart-deck-10	D3	5	2	1	23	19	42
kart-deck-11	D3	5	1	0	23	19	42
kart-deck-12	D3	5	1	0	21	15	68
kart-deck-13	D4	5	4	0	33	23	403
kart-deck-14	D4	5	4	0	33	23	392
airframe-01	D1	21	0	0	21	21	21
airframe-02	D1	21	0	0	33	33	33
airframe-03	D2	8	0	0	16	14	16
airframe-04	D2	8	0	0	16	14	16

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
airframe-05	D2	8	0	0	17	13	30
airframe-06	D2	8	0	0	17	13	30
airframe-07	D3	8	2	0	22	20	22
airframe-08	D3	8	2	0	22	20	22
airframe-09	D3	8	2	1	9	9	9
airframe-10	D3	8	2	1	9	9	9
airframe-11	D3	8	1	0	9	9	9
airframe-12	D3	8	1	0	9	9	9
airframe-13	D3	8	4	0	26	24	26
airframe-14	D3	8	4	0	26	24	26
truck-cab-01	D1	18	0	0	18	18	18
truck-cab-02	D1	18	0	0	27	27	27
truck-cab-03	D3	12	2	0	40	34	1522
truck-cab-04	D3	12	2	0	36	30	809
truck-cab-05	D3	12	2	1	21	19	22
truck-cab-06	D3	12	2	1	21	19	22
truck-cab-07	D3	12	1	0	21	19	22
truck-cab-08	D3	12	1	0	21	19	22
truck-cab-09	D4	12	4	0	42	36	1546
truck-cab-10	D4	12	4	0	38	32	831
house-roof-01	D1	18	0	0	18	18	18
house-roof-02	D1	18	0	0	22	22	22
house-roof-03	D2	5	0	0	21	17	37
house-roof-04	D2	5	0	0	21	17	37
house-roof-05	D2	5	0	0	22	18	45
house-roof-06	D2	5	0	0	22	18	45
house-roof-07	D3	5	2	0	27	23	54
house-roof-08	D3	5	2	0	27	23	50
house-roof-09	D3	5	2	1	17	17	19
house-roof-10	D3	5	2	1	17	17	23
house-roof-11	D3	5	1	0	17	17	23
house-roof-12	D3	5	1	0	17	17	19
house-roof-13	D3	5	4	0	33	29	76
house-roof-14	D3	5	4	0	31	27	68
sports-body-01	D1	22	0	0	10	10	10
sports-body-02	D1	22	0	0	11	11	11
sports-body-03	D1	8	2	0	17	17	17